

Artificial Intelligence

Biological Neural Networks

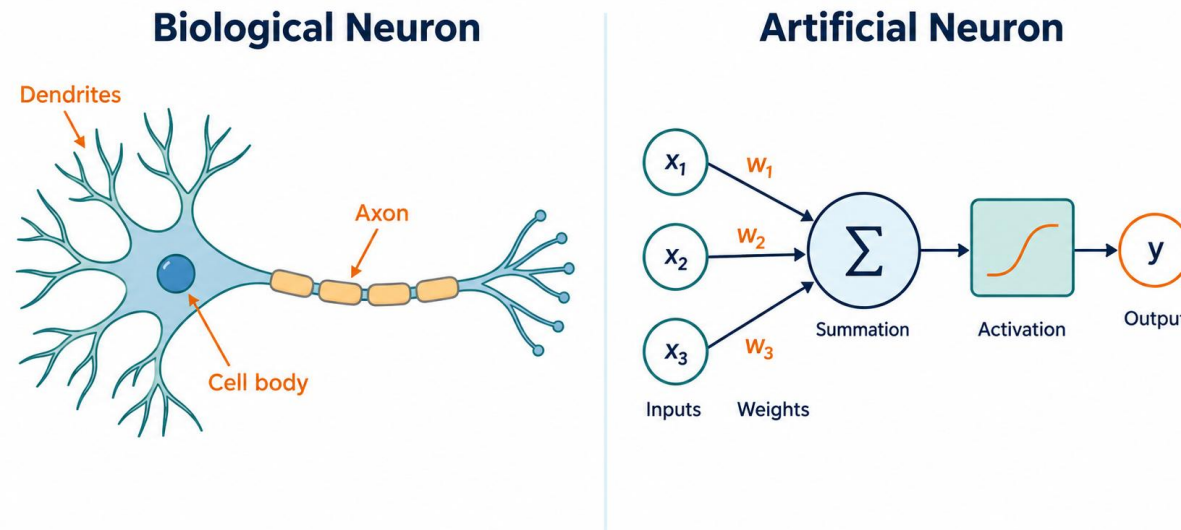
Lecture 03

The Islamic University in Najaf Al-Ashraf
College of Technical Engineering
Department of Computer Techniques Engineering
Instructor Jaafar Al-Kafnawiz

Najaf Al-Ashraf



Biological and artificial neurons



Inputs, weights and activations provide a simplified computational analogy.

Biological nervous system



The nervous system processes signals using connected neurons. Each neuron receives signals, integrates them, and may generate an output.

Learning changes the strength of connections between neurons.

Structure of a biological neuron



Dendrites receive signals. The cell body integrates signals. The axon transmits the response. Synapses connect one neuron to another.

Input through dendrites; integration in soma; output through axon.

Synaptic transmission



A presynaptic neuron releases a signal. The synapse influences the postsynaptic neuron. Excitatory inputs increase activation.

Inhibitory inputs reduce activation.

Biological and artificial neurons



Dendrites correspond to inputs, and synaptic strengths correspond to weights.

The cell body motivates input summation. Firing motivates an activation function.

The axon corresponds to the output passed to other units.

Artificial model: $\text{net} = \sum_i w_i x_i + b$, followed by $y = f(\text{net})$.

This correspondence is a simplified computational analogy.

Worked neuron calculation



Inputs: $x = [1, 0, 1]$. Weights: $w = [0.6, -0.4, 0.8]$. Bias: $b = -0.7$. Net input = $0.6(1) - 0.4(0) + 0.8(1) - 0.7 = 0.7$.

With step activation, $y = 1$ because $\text{net} \geq 0$.

Threshold activation



The output changes abruptly at a threshold. Useful for binary logic models.

The derivative is zero or undefined, so gradient training is difficult.
 $f(\text{net}) = 1$ if $\text{net} \geq 0$; otherwise 0.

Sign and linear activations



Sign: output is +1 for $\text{net} \geq 0$ and -1 otherwise.

Sign activation represents bipolar decisions in early neural models.

Linear: output equals the net input, $f(\text{net}) = \text{net}$.

Linear outputs suit numerical prediction. Stacking only linear layers still gives a linear mapping.

Sigmoid activation



Maps any real input to a value between 0 and 1. Smooth and differentiable. Useful for binary probability output.

Large positive or negative inputs produce small gradients. $\sigma(z) = 1 / (1 + e^{-z})$.

Hyperbolic tangent



Maps input to the range -1 to 1. Zero-centered output often helps optimization.

Can still suffer from saturation. $\tanh(z) = (e^z - e^{-z}) / (e^z + e^{-z})$.

Activation function comparison



Step and sign produce discrete decisions for threshold models.

Linear activation retains the net input for numerical outputs.

Sigmoid maps a scalar to $(0, 1)$, while tanh maps it to $(-1, 1)$.

$\text{ReLU}(z) = \max(0, z)$ introduces a simple nonlinearity.

Softmax converts a vector of class scores into probabilities that sum to one.

Early neural structures



McCulloch and Pitts introduced a logical neuron model. Hebbian learning linked co-activation with stronger connections.

The Perceptron introduced automatic weight learning. Adaline used a linear output and an error-based rule.

Network as a directed graph



Neurons form nodes. Weighted connections form directed edges.
Layers organize information flow.

Cycles distinguish recurrent networks from feedforward networks.

Biological and artificial comparison



Biological neurons communicate through complex electrochemical processes.

Artificial neurons represent inputs as numbers and connection strengths as weights.

Biological plasticity motivates learning, while artificial training applies mathematical update rules.

The artificial model is useful for computation but does not reproduce the full biological system.

Benefits and limitations of neuron models



Benefit: weights and activations provide a simple model of input integration.

Benefit: nonlinear activations let multilayer networks model complex relationships.

Limitation: threshold functions do not provide useful ordinary gradients.

Risk: sigmoid and tanh can saturate. ReLU units may remain inactive.

Artificial neurons omit many biological processes.

Activation functions: comparison



Function	Output range	Typical role
Sigmoid	0 to 1	Binary output probability
Tanh	-1 to +1	Centered hidden response
ReLU	0 and positive values	Nonlinear hidden layers